

BREAKING GROUND

*(in-scene rewrite — companion draft,
staying inside her head)*

They were still on the observation gallery when the first work order came in, hours after the boards had gone dark, hours after the corridor outside had emptied of everyone but the two of them and a kind of exhaustion that only shows up once you've been handed something too large to carry to bed.

Ilse read the anonymous message again on the private channel. *You are going to build something and then discover you cannot control*

what it becomes. She still hadn't answered it. She never would, not publicly. She read it a second time in the gray hour before the station called it morning, and somewhere in the reading it stopped being a warning about a future machine and became, instead, a description of the exact morning she was standing in.

Nobody had thought to ask who was allowed to spend the money.

She didn't notice that part until the second morning — the first had gone entirely to the disorientation of having won. Four hours of sleep in a supply berth neither of them had used in a year. A steward's

knock. A signature Ilse gave before she'd finished reading what she was signing, because that morning, saying yes to paperwork felt like the easiest yes she'd ever be asked for again.

Perrin found her in the mess hall doorway, tablet held slightly away from her body, the particular way a person holds something they already know is about to ruin someone's day.

"The Council ratified a mandate," Perrin said. "It didn't ratify an org chart. There's a line that says who oversees this — plural, five settlements deep. There's a line about what it can and can't be used

for. There's no line that says whose signature authorizes a procurement order past a certain threshold, and that threshold shows up in about eleven days, when the first contractor asks who to bill."

Ilse looked up from the tablet she'd signed without reading. "I thought that was implied."

"Implied by what?"

"By the fact that I proposed it."

"You proposed a theory," Perrin said, flat, not unkind — eleven months in the Council's machinery had worn the kindness down to something more useful than kindness. "A theory doesn't sign contracts. You had a mess hall and a whiteboard. It

worked because the theory was strong enough to carry itself through a hearing. It will not carry itself through a procurement dispute."

Corin, three seats down, had stopped eating. "So who is."

"Nobody. Currently." Perrin set the tablet down at last. "The Council assumed you'd already have a university or a settlement government or a corporate division behind you before the vote ever happened. You had eleven months of drafting and a proof strong enough to survive a hearing. That's not the same thing."

Ilse set the tablet down too. Three years of watching a simulation return a number she hadn't expected had taught her the exact shape of this feeling — a floor she'd assumed was load-bearing, turning out to be assumed rather than verified.

"Then we build the structure," she said.

"You build it in eleven days," Perrin said, "or the first contractor walks, and every contractor after them hears why."

Perrin spent four of the eleven days arguing with a Council registrar over a legal category that had existed on

paper for eight years and never once been used — *provisional charter trust* — built for exactly this shape of problem and never tested because nothing had ever needed it.

"You're the first thing that's ever needed this category," Perrin told them on the fifth day, in a voice that might have been satisfaction and might have been exhaustion wearing satisfaction's clothes. "The registrar had to look it up. I don't think they believed it was real until I sent the statute number."

Ilse became, on paper, the trust's accountable executive — her name on every document large enough to need one, her name the place any

real legal exposure would land,
rather than dissolving into an
institution big enough to absorb it.
Corin became technical director,
less legal weight, more actual
authority over the engineering.
Neither of them had planned it that
way. It felt, once it existed, like the
only honest arrangement available.

"You realize what this makes you,"
Corin said, on the eighth day,
reading the incorporation papers in
the same mess hall that had hosted
every important conversation this
project had ever had.

"What."

"The single most legally exposed
person in this quadrant. Over a

project that doesn't have a building yet."

"I've been the most exposed person in this quadrant since run four hundred and twelve," Ilse said. "This just made it official."

Tavik, reading over both their shoulders, didn't say anything for a while.

"Verdent had an accountable executive," he said finally. "Took the Council four years to determine who, because the authority had been diffused across three firms until nobody could be made to answer for it. Nobody went to prison. Nobody lost a license. A hundred and forty thousand people

died, and the closest thing to accountability was a joint statement from a consortium that dissolved itself eighteen months later, specifically so it couldn't be sued." He looked at her a long moment. "I would rather you carry this than watch it disappear into that kind of fog. I mean that as something closer to a gift than a warning. I understand it won't feel like one most days."

The first test came on the ninth day, before the paperwork had even cleared final review — a prefab-housing contractor missed a delivery window by six days, and when logistics called to ask why,

simply stopped answering the channel.

"They think there's nobody to sue," Perrin said, within the hour. "They've done it before, to smaller outfits. Miss a delivery, absorb the penalty, move on before anyone's assembled the standing to pursue it."

"Does it work against us."

"Not if I file today, under your name, citing the trust's registered standing." Perrin looked at her steadily. "I want you to understand what that means. Your name, on a public dispute, on this project's ninth day of legal existence. If we

lose, it costs us standing with every contractor watching. If we win —"

"File it," Ilse said, and didn't sleep well that night — not doubt about the decision, just the still-unfamiliar weight of watching her own name become an instrument other people would use to measure whether any of this was real.

They won in eleven days. Perrin brought her the ruling in person.

"Word travels fast in that industry," Perrin said. "I wish I could tell you that's the last time you'll need me to weaponize your own name on your behalf. It won't be. It's just the first time, and I wanted you to see one

go well before the harder ones start."

The site search had been sitting in a subclause since the ninth draft of the charter, and Ilse hadn't thought seriously about it until someone finally had the legal standing to purchase a survey. Eleven candidates. Nine rejected in the first pass — unclaimed didn't mean unwanted. Two marginal, worlds with breathable air or workable soil that some settlement was quietly planning to claim once its own situation stabilized.

"We're not doing that," Ilse said, before Corin finished the sentence

out loud. "Whatever this costs, it isn't costing someone a world they were going to live on."

She walked the surface of one candidate herself, because she didn't trust herself to reject it on paper without standing on it first. A settlement liaison met her there, told her, with the specific pride of a man who'd spent years angling for this, about the families his people hoped to relocate here within the decade.

She didn't tell him what she was actually evaluating. She thanked him for the tour. The liaison had shown her, on a tablet he clearly wasn't supposed to have brought

along, a rough housing sketch with a school penciled into the northeast corner and a note beside it, still unofficial, that just said *room to grow*. She filed the world unsuitable that evening. She thought about the school for three weeks afterward, longer than she thought about the mineral report.

That left the dead moon. No atmosphere, no habitable neighbors, gravity thin enough to complicate construction without ruling it out, geology stable enough to hold real mass. Nobody had ever filed interest in it, because there was nothing on it worth wanting.

"It's perfect," Corin said, at the fourth review, and then, catching how the word landed in the room: "I don't mean it the way it sounds."

"I know what you mean," Ilse said.

Nobody said the other thing out loud, not that session, not for two more weeks. Ilse went back to the survey data alone that night anyway, past the mineral tables and the gravity readings, to a section near the back nobody had actually asked her to read — projected failure radius, worst case, total containment loss. She read the number twice. She didn't write anything in the margin. She just sat with it long enough that when she

finally closed the file, her hand had gone stiff from not moving.

She said it out loud eventually, to Tavik, three weeks into the survey, walking the perimeter in a suit rated for an atmosphere this moon didn't have.

"We picked it because nobody wanted it," she said. "That's the official reason. I keep thinking about the other one. The one nobody's put in a report yet."

"Say it."

"If this goes wrong the way Verdent went wrong, it goes wrong *here*. No population. I picked a place to fail safely before I've even built the thing that might fail." She stopped

walking. "I don't know what it says about me that I did the arithmetic on our own catastrophe before we'd laid a single foundation."

Tavik's boots kept crunching against regolith that had never felt a footstep before that week — ground so undisturbed their own prints would likely outlast everything they were about to build on top of them.

"It says you learned the actual lesson of Verdent," he said. "Not the one people usually take — *don't attempt things that might fail*. The lesson is: if you're going to attempt something that might fail, fail somewhere it doesn't cost you a world. I spent forty years wishing

someone had done that arithmetic before the first settlers moved onto Verdent's first continent. You did it before you'd poured a single foundation." A pause, longer than the silence needed. "That's the first time in this whole project I've watched you get ahead of a mistake instead of answering for one that already happened."

Ground broke on a Tuesday, by a calendar that meant less with no day-night cycle worth the name.

The regolith took the first module's footing with a small, undramatic shudder that traveled up through the observation deck and into the

soles of Ilse's boots — the first physical thing this project had produced in nine months of theory and law and argument. A second module followed within the hour. A third before the transport had finished unloading. By shift's end, six modules sat where nothing had ever sat, in the loose grid the site plan optimistically called a village.

One new arrival crouched near the second module's entrance, still in transit gear, and pressed a bare gloved hand flat against ground that had never held anyone's weight. He held it there a few seconds longer than the gesture needed, then pulled it back and studied the dust caught in the glove's seams like it

might tell him something. The next one out of the module walked a half-step around the print rather than over it, and didn't appear to notice he'd done it.

Ilse had expected to feel triumph. What she felt instead was closer to vertigo.

"You look like you're going to be sick," Corin said, arriving with two cups of something that was technically coffee.

"I keep doing the math on how many people are down there," Ilse said. "Two hundred, this week. Four thousand within the year. Twelve thousand within three. I have never

managed more than the six people in that mess hall."

"The mathematics was the hardest part you could do alone," Corin said.

"This is the hardest part that requires everyone else, and you're not doing it alone, because you already built the structure that makes sure you can't." She didn't look at Ilse when she said the next part. "Perrin didn't spend eleven days inventing a legal category so you could carry twelve thousand people by yourself. You didn't carry the theory alone either, for what it's worth. You just remember it that way because you were the one standing at the podium."

A procurement request crossed Ilse's desk in the third month, and she almost approved it without reading past the line item: *neonatal environmental unit, x1, priority routing*.

She read it twice, assumed it belonged to Corin's containment research, some component she hadn't been briefed on, and pulled the manifest to check which team had requested it.

Logistics. Not engineering.

She found Perrin in the corridor outside the allocation office. "What is this."

Perrin didn't glance at the screen Ilse turned toward her. "Pressure-

rated infant care unit. Third construction shift. Someone's seven months along — arrived that way, didn't say anything until medical intake flagged it."

"We don't have a policy for this."

"No," Perrin agreed. "We have a policy for engineers and contractors and logistics staff. Nobody wrote one for what happens when engineers and contractors and logistics staff start having actual lives out here, because nobody building a research platform usually has to. I ordered two units instead of one."

"Why two."

"Because this will not be the last requisition of this kind I file," Perrin said, "and it's faster to have the second one already in transit than to explain, a second time, why an engineering project needs something this small and this permanent."

The power problem finally got said out loud in month five, and it was Corin who said it, in the silence Ilse had learned to recognize — the same silence from the mess hall, eighteen months and a lifetime ago, right before the third hole turned out not to be a hole at all.

"We don't actually know how to build this yet," Corin said. "Not the containment. The power underneath it. Every other problem on this table is a hard problem in a known category — hiring, logistics, legal structure, things humanity's solved before at smaller scale. The power problem isn't in a known category. I've spent a month reading every energy archive I have clearance for, and I keep hitting the same wall: everything that's been tried at anything near this scale was tried, and it didn't work, and it got quietly wound down a generation ago, because nobody's needed it since. Until now."

She'd watched one of those lines die herself, three weeks earlier, standing behind a containment window while a junior researcher named Halden ran what everyone already suspected would be the last test on it. Plates a few nanometers apart. Instruments calibrated for a yield too small for anything but the most sensitive detection anyone had. For eleven seconds, it worked exactly the way the theory said it should.

"There it is," Halden said quietly, watching the readout. "It's real. It's always been real."

"How much," Corin asked, already knowing, asking anyway because

someone had to say the number.

"Enough to power a light for about four seconds. We need nine orders of magnitude beyond where this rig sits, and I don't think nine orders of magnitude is a distance any number of decades closes with this mechanism. The effect isn't the problem. I think it might be a wall built out of the actual shape of the physics."

Nobody argued with him. Corin closed the file that afternoon. She told Ilse that evening that closing a real, honestly tested line felt worse than the mess hall's original uncertainty ever had, because

uncertainty had at least still had the shape of hope in it.

"I don't have an answer for you tonight," Ilse said.

"I know. I'm asking you to sit with the fact that I don't currently believe the answer exists in anything either of us already knows how to build."

Corin's voice didn't do anything dramatic on the next part, which was somehow worse than if it had.

"I think, for the first time since the mess hall, I'm genuinely afraid we scoped a promise we can't keep."

Ilse thought of the charter clause Perrin had fought hardest for — bound to honesty even when the honest answer was uncomfortable

— and understood it had been written for exactly this moment. Not for some hearing five years off that neither of them could picture yet. For this: two people in front of a whiteboard at an hour they'd both stopped tracking, admitting to each other, before admitting it to anyone with the standing to stop them, that they did not currently know how to build the thing they'd promised a room full of strangers they could build.

"Then we say that," Ilse said. "Not to the Council yet — there's nothing to report but an open problem, and an open problem isn't a failure, it's just research before it's finished being research. But we say it to each

other, the way you said it to me in the mess hall. We don't have an answer. We have a wall. I would rather stand in front of a wall honestly than pretend to either of us it's a door we simply haven't opened yet."

Perrin found them both on the observation platform some nights later, carrying a tablet she didn't immediately offer, which had become its own kind of warning.

"The oversight committee's asked for its first formal briefing," she said. "Not hostile. Curious — the kind that turns hostile if it goes unanswered long enough. Four thousand

workers, a footprint visible from orbit, and nothing built yet that looks like the machine the charter promised them."

"Tell them the truth," Ilse said. "We don't have the power architecture yet. We're looking. We haven't found it."

"I will tell them exactly that," Perrin said. "I want you to understand what that honesty is going to cost, going forward, the same way I wanted you to understand what your name on a contract dispute would cost. Some of them will respect it. Some of them will hear an open question about the central mechanism and start asking

whether the resource allocation was premature. I don't have a way to guarantee which reaction wins."

"Then that's the cost," Ilse said. "We agreed to it in the hearing before we agreed to anything else."

Neither of them moved from the platform for a long time, watching four thousand lights scattered across ground that had never held a light before this year — one of them, she knew now, belonging to someone who hadn't come there to build anything at all, who had simply begun there — staked on a promise still waiting for its oldest, plainest question to find an answer.

*How do you power a machine
nobody has ever needed to build
before, using physics nobody has
ever needed to push this far.*

Ilse thought of her sister's transit route, flown safely a few hundred times before the hundred and first crossing found the one trajectory nobody had modeled precisely enough to catch. She had built this entire project on the conviction that precision was a thing you could afford, if you were willing to pay for it before the cost came due instead of after. She had not seriously considered, until tonight, what it would mean if precision itself turned out to be waiting on a piece of physics that didn't exist yet — if

the thing standing between four thousand people's trust and the promise she'd made them wasn't a matter of resolve at all, but a fact about the universe nobody currently alive knew how to change.

She didn't know if that made the problem harder, or just older than she'd understood it to be.

She only knew she wasn't going to stop looking. She hadn't stopped at run four hundred and eleven.

Stopping had never once, in her whole life, been the shape grief actually took.

Chapter two, all four primary targets fixed the way the sheet asked —

*night realization now incomplete
(Tavik still gets to be the one who
says it), the school sketch replacing
the "whole reason" tail, the
glove/dust/footstep replacing the
touch-interpretation, and the honesty
clause now grounded in the
whiteboard instead of
foreshadowing Atlas's interiority.
Added the procurement/baby beat
where the sheet suggested it,
underplayed, two units instead of
one. Tell me what's still lying there.*